

WILLIAMSPORT, PA, USA

WMO: 725140

Lat: 41.243N

Lon: 76.922W

Elev: 520

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14778

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.3	(c) 10.1	(d) -6.9	(e) 3.9	(f) 8.8	(g) -2.2	(h) 5.0	(i) 13.4	(j) 24.5	(k) 31.5	(l) 22.1	(m) 30.9	(n) 6.7	(o) 270	(p) 0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.8	(c) 90.7	(d) 72.4	(e) 87.7	(f) 71.2	(g) 85.0	(h) 69.9	(i) 75.5	(j) 85.5	(k) 74.1	(l) 83.1	(m) 72.7	(n) 80.9	(o) 8.6	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	123.0	80.1	71.3	117.7	78.6	70.0	112.6	77.1	39.3	85.8	37.9	83.1	36.6	80.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	18.2	16.2	DB	-1.0	95.8	6.7	3.1	-5.8	98.0	-9.7	99.8	-13.4	101.6	-18.3	103.8	
			WB	-2.0	78.4	6.4	1.5	-6.5	79.4	-10.2	80.3	-13.8	81.1	-18.4	82.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	27.3	30.1	38.8	50.6	60.8	69.5	73.6	71.9	65.0	53.3	42.0	33.0
	DBStd	17.95	10.10	9.17	9.49	8.46	7.76	6.06	5.07	5.00	6.68	7.75	8.07	8.39
	HDD50	2576	705	558	367	92	6	0	0	0	0	53	263	532
	HDD65	5749	1169	977	812	437	181	27	2	6	82	373	690	993
	CDD50	3100	1	1	20	110	341	584	732	680	450	155	23	3
	CDD65	798	0	0	0	6	51	161	268	221	81	10	0	0
	CDH74	6616	0	0	14	133	534	1375	2293	1653	558	56	0	0
	CDH80	2149	0	0	3	36	165	429	836	520	152	8	0	0
Wind	WSAvg	6.0	7.2	7.2	7.5	7.3	6.0	5.3	4.8	4.3	4.6	5.3	6.1	6.7
Precipitation	PrecAvg	40.9	2.7	2.5	3.2	3.3	3.8	4.0	4.1	3.5	4.0	3.2	3.7	2.9
	PrecMax	59.2	8.3	8.4	5.9	7.6	7.3	16.1	9.6	7.7	12.6	9.6	7.8	7.0
	PrecMin	30.1	0.5	0.6	0.9	0.7	0.8	0.6	1.2	0.9	0.5	0.2	0.8	0.7
	PrecStd	7.0	1.7	1.6	1.3	1.8	1.7	2.4	1.8	1.6	2.7	2.0	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	61.3	74.2	84.7	89.4	92.2	95.4	93.7	89.6	80.1	68.3	62.3
		MCWB	56.3	50.2	57.4	63.5	69.4	73.7	74.8	73.8	71.6	67.6	58.4	57.4
	2%	DB	50.3	51.6	65.3	77.6	84.7	88.3	91.1	88.8	84.3	74.2	63.5	53.5
		MCWB	45.7	43.3	52.2	59.6	67.2	71.4	73.0	71.8	69.7	64.6	56.1	49.1
	5%	DB	44.6	46.8	59.0	71.4	80.2	84.8	87.9	85.4	80.2	70.3	59.2	48.8
		MCWB	40.7	40.3	49.4	56.2	65.2	70.0	71.8	70.6	67.9	61.7	52.2	44.8
	10%	DB	40.7	42.6	53.7	66.4	75.4	81.8	84.7	82.4	76.4	66.3	55.3	44.7
		MCWB	37.3	37.8	46.0	52.9	62.8	68.4	70.5	69.5	66.0	58.9	49.3	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	52.8	60.3	66.5	72.9	76.3	78.1	77.3	75.1	70.5	61.8	57.7
		MCDB	58.9	58.6	71.1	79.6	84.6	86.7	89.7	88.0	84.1	77.0	65.9	61.3
	2%	WB	46.3	45.2	55.0	62.3	69.8	74.2	75.9	75.0	72.5	67.1	57.7	50.5
		MCDB	48.8	49.7	62.5	72.7	80.1	83.9	86.4	83.8	80.0	72.3	61.6	52.8
	5%	WB	41.2	41.4	50.5	59.0	67.7	72.5	74.4	73.3	70.6	63.6	53.8	45.5
		MCDB	44.2	46.0	57.6	68.2	76.9	81.4	83.6	81.2	76.6	68.7	57.7	48.3
	10%	WB	37.4	37.9	46.2	55.3	65.1	70.7	72.8	71.7	68.6	59.9	50.2	41.4
		MCDB	40.2	42.2	53.0	64.1	72.7	78.7	80.9	78.8	74.0	64.9	54.6	44.7
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.9	18.2	21.7	21.6	20.4	20.8	19.9	19.8	19.0	16.1	12.8
		MCDBR	18.2	22.1	27.1	30.7	28.0	24.4	23.7	23.1	24.0	24.1	22.5	17.3
		MCWBR	15.1	15.6	16.3	16.1	13.3	10.9	9.4	9.7	11.3	13.9	15.3	14.2
	5% WB	MCDBR	17.3	19.2	24.5	25.9	23.6	20.7	20.1	19.3	19.6	19.9	18.7	15.2
		MCWBR	15.1	15.0	16.5	15.9	12.0	10.4	9.1	9.1	10.3	12.6	15.1	14.0
Clear-Sky Solar Irradiance	taub	0.316	0.329	0.364	0.388	0.422	0.478	0.486	0.477	0.390	0.372	0.329	0.309	
	taud	2.372	2.325	2.318	2.302	2.236	2.112	2.105	2.113	2.340	2.390	2.474	2.450	
	Ebn at Noon	263	278	280	280	272	256	252	250	268	259	258	258	
	Edh at Noon	26	32	36	40	43	49	49	48	36	30	24	23	
All-Sky Solar Radiation	RadAvg	523	802	1114	1436	1637	1779	1814	1596	1296	877	595	428	
	RadStd	40	93	97	156	175	156	116	97	166	101	73	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46) Station Only		(d) +0.92	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) +221	(l) N/A	(46)
(47) Regional (38 neighbors)		(d) +0.72	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) +198	(l) N/A	(47)

Nomenclature: See separate page